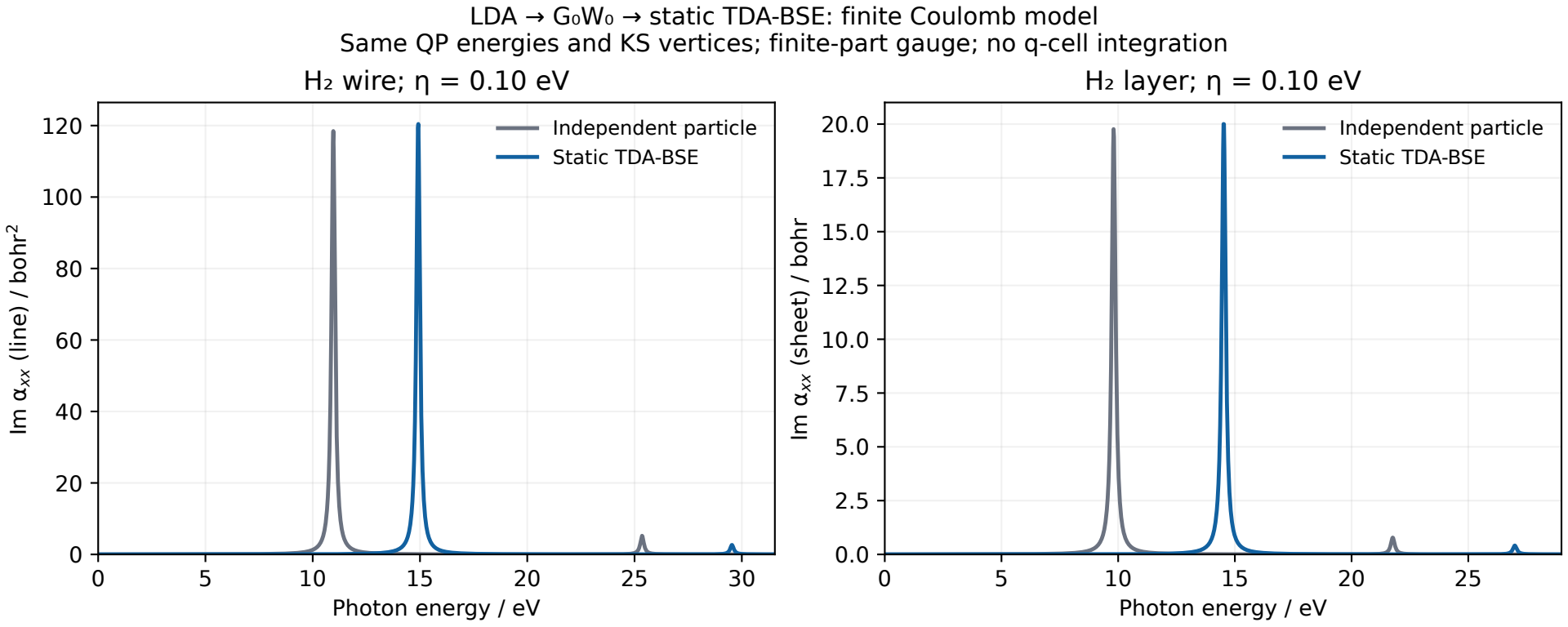
LDA → G₀W₀ → static TDA-BSE: finite Coulomb model
Same QP energies and KS vertices; finite-part gauge; no q-cell integration
H₂ wire; η = 0.10 eV
H₂ layer; η = 0.10 eV
Independent particle
Static TDA-BSE
Im α_xx (line) / bohr²
Im α_xx (sheet) / bohr
Photon energy / eV
0
5
10
15
20
25
30
0
20
40
60
80
100
120
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0